

Feasibility of Language Detection Attacks on LLM-generated Random Sequences in Federated Learning Environments

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Section A: Literature Review

Paper 1: "A Survey on LLM-Generated Text Detection: Necessity, Methods, and Future Directions"

Venue: Association of Computational Linguistics (ACL) 2025

Summary: This paper addresses detection (identifying content generated by llms), attribution (identifying which llm generated the content), and characterization (determining the intention of the generated content - i.e., what prompt was used to generate the content).

Strengths: Many of the techniques used for binary classification between whether text is human or llm generated can be tested to classify whether random number generation is prompted through english or chinese. The paper also introduces many benchmarks to use for llm-generated text detection.

Weaknesses: Much of the detection involves using features of grammar and emotional intensity, which simply generating random numbers does not have. However, different cultures may have different emotional intensity for certain numbers, so perhaps this can contribute in a difference of frequency or pattern in the random number generation that mimics that of language. For uniform random number sampling we can use a baseline from mathematical concepts rather than the text benchmarks.

Paper 2: "Defeating Nondeterminism in LLM Inference"

Venue: Thinking Machines blog (Sep 10, 2025)

Summary: The paper goes beyond addressing floating-point non-associativity as the culprit of the nondeterminism in llms. Floating-point non-associativity is just that when floating-point numbers are added together in a different order, the non-associativity gives us a different result due to dropping in precision. The paper comes to the conclusion that the primary reason for the nondeterminism is that the batch-size nondeterministically varies. The author proposes a method for batch invariance for determinism.

Strengths: It directly introduces methods for how toggle determinism. Moreover, the nondeterminism applies not only to GPUs, but also CPUs and TPUs.

Weaknesses: To what end this allows for more performant uniform random number sampling is unknown.

Paper 3: "Can LLMs Generate Random Numbers? Evaluating LLM Sampling in Controlled Domains"

Venue: ICML 2023

Summary: The paper broadly addresses the question of how well an llm performs when replicating a distribution. The paper categorizes llm sampling approaches into non-autoregressive sampling (NARS) and autoregressive sampling (ARS). NARS is when a fixed prompt is used to generate many samples independently. ARS is when a fixed prompt is used to generate a sample, and following samples are generated autoregressively. These sampling techniques are evaluated in control domains of uniform random number sampling (URNS) and probabilistic context-free grammar (PCFG).

Strengths: It directly addresses the ability of llms to perform random number generation. URNS performed better than PCFG, though both are not adequate. Instruction fine-tuning and examples improved the performance, however.

Weaknesses: The language of the prompts used were not specified. The random number generation also occurred within the range $[0, 1]$ as floating point, which may have other biases compared to integers. Moreover, they did not evaluate tokenization and training data biases.

Paper 4: "Deep Leakage from Gradients"

Venue: NeurIPS 2019

Summary: This is a foundational paper that demonstrates techniques of reconstructing training data and labels from weights during federated learning. They demonstrated that the batch size and model depth also affects deep leakage. The paper gives a framework for gradient-based privacy attacks that we can adapt our data.

Strengths: The paper focuses on supervised learning tasks with image-label pairs and language tasks, providing a framework for gradient-based privacy attacks.

Weaknesses: We have a different set of data that we may have to create different success metrics for, as the paper's framework may not directly apply to our random number generation scenario.

Section B: Project Description

Project Overview and Problem Statement

The main conclusion we want to come to for this paper is to find how accurately we can categorize the prompt language used for the generation from federated learning. The problem statement is to determine whether prompting with different languages, like english or chinese, leads to different distributions in the sequence of random integers, and to what extent we can make the model more deterministic to hide these differences.

Relevance to Course: This project explores linguistic/cultural fingerprinting through generated content in a federated learning context, investigating whether language-specific biases in LLM outputs could serve as identifying characteristics that persist even in distributed learning scenarios. This relates to behavioral biometrics and privacy attacks in federated systems where gradients may leak information about the prompting context.

Nature of the Project: "Test feasibility of language detection attacks on LLM-generated random sequences in federated learning environments"

Relationship to Literature

The primary use of the survey for detecting llm-generated text would be to modify linguistics methods of categorization to random number generation. For instance, using techniques for characterization, we could potentially identify what model is being used to generate the numbers.

The "defeating nondeterminism in llm inference" applies to the research project because we could use the research ideas to adjust the randomness for the llm when prompted to generate a sequence of random numbers.

The "can llms generate random numbers" paper directly addresses our problem of random number generation. We can use the techniques used in this paper to analyze the distributions, and use them as features to design our own architecture to classify whether the prompt was in english or chinese.

The "deep leakage from gradients" paper provides a framework for how deep leakage can occur, and demonstrates how gradient pruning can be used to mitigate this. The paper presents an optimization algorithm that does not depend on generative models.

Key Modules/Components

Key modules of the project are the lstm, federated learning architecture, language model, the prompt language, the length of the sequence of random integers, and the range of the integers. The key components can be structured as:

1. Data Generation Module: LLM interface for generating random sequences with different language prompts
2. LSTM Processing Module: Feed the sequences into lstms for pattern recognition
3. Federated Learning Architecture: Distributed learning system for training detection models
4. Statistical Analysis Module: Use nist test, autocorrelation analysis, and spectral analysis to determine whether there's differences between the distributions and also determine the randomness of the sequences
5. Feature Engineering Module: Determine features to use for the federated machine learning model
6. Model Refinement Module: Tweak the language model for greater or less randomness - can tweak the model in different ways (e.g., maybe the model is tweaked to be more deterministic, but the model generates a single distribution that is more random)

While we can scope the project to these modules, if time allows we can expand our scope to consider a greater variety of language models and prompt languages. We can also determine whether the accuracy of our detection model changes depending on the length of the sequence, multiple generations of a fixed n-length sequence, the range, and switching to floating-point numbers. Moreover, we may also be able to identify the language model used to generate the sequence of numbers.

Timeline

- Weeks 1-2: Find two language models to experiment with - find existing open source models that can be run locally
- Weeks 3-4: Generate a dataset with the language model - use chinese and english to generate random integers, generate sequences of random integers of different sizes (i.e. 5-length to 20-length)
- Weeks 5-6: Use statistical techniques to determine whether there's differences between the distributions - use nist test, autocorrelation analysis, and spectral analysis, also determine the randomness of the sequences
- Weeks 7-8: Feed the sequences into lstms and determine features to use for the federated machine learning model
- Weeks 9-10: Tweak the language model for greater or less randomness - with every tweak, we can perform the above steps in the timeline, maybe automate the above steps in a pipeline
- Weeks 11-12: If time allows, use floating-point to determine if there's a difference there
- Weeks 13-14: Results analysis, report writing, presentation preparation

Dataset Details

The dataset used for the project will be synthetically generated. We can also use datasets of pseudo-random numbers as a baseline to compare to.

Dataset Specifications:

- Primary: Synthetically generated sequences using 2+ open-source LLMs
- Sequence lengths: 5-length to 20-length sequences
- Languages: Chinese and English prompts for generating random integers
- Baseline: Mathematically generated pseudo-random sequences for comparison

Limitations and Considerations: While there are datasets where human participants randomly generate sequences of numbers within a range, these are primarily in western institutions and do not account for cultural diversity. For instance, we can use the dataset from Wagenaar, though these participants are primarily Dutch. One possibility is to combine datasets to represent populations proportionally, but this is probably too extensive given the single-semester timeline. To better distinguish between an english and chinese prompt to generate random numbers, having human-generated numbers from both english and chinese speakers to train on could be useful. But we were not able to find a chinese dataset. Maybe a greater influence is the geographical location the participants have lived in, perhaps better distinguishing between eastern and western cultural influence on random number generation.

Scope and Limitations:

- Focus on binary classification (English vs Chinese) in federated learning context
- Limited to integer sequences initially, with potential expansion to floating-point
- Constrained by computational resources for local LLM execution and federated learning simulation

- Cultural bias detection rather than individual identification, with emphasis on gradient leakage scenarios